

---

---

# Bayer & myHealthteam

— MHT Menopause Community —  
Review - Climara

---

---

# AGENDA

- Approach
- Solution Overview
- Insights and Findings
- Additional Considerations & Next Steps
- Q & A
- Appendix

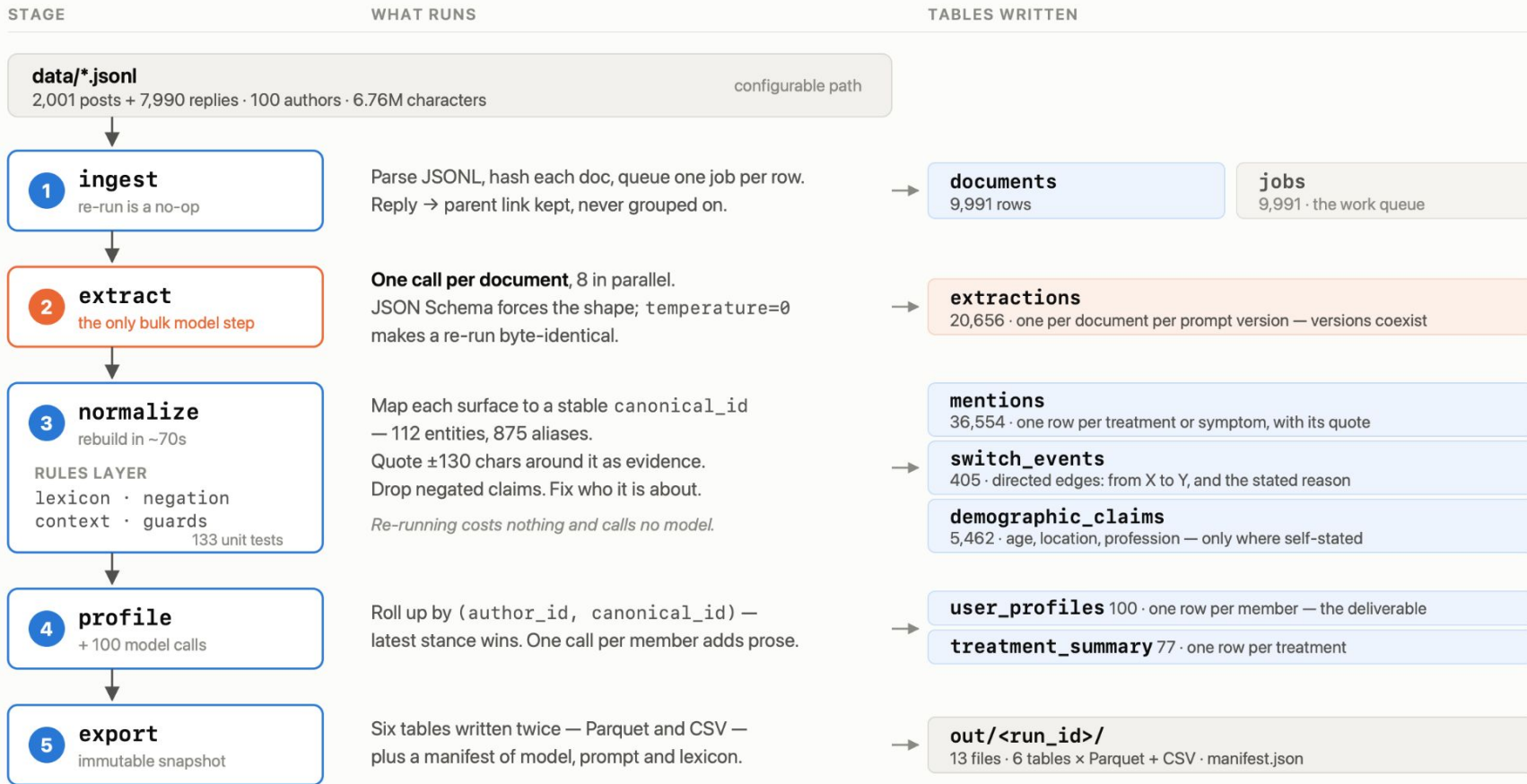
# Approach

- **Goal:** Extract a set of analytical data products to support Climara patient community analyses
- Data
  - 2 [\*.jsonl] files with ~10,000 MyHealthTeam Menopause Community Posts and Replies
    - ~2,000 posts
    - ~8,000 replies (\*to provided parent posts)
  - Included data attributes: Post/Reply - body, author, unique identifier and creation date
- Solution
  - Overview

# Forum text → a per-member structured dataset

model deterministic

One model call per document. Everything after it is deterministic and re-runnable for free.



# Approach

- Assumptions/Caveats
  - User-generated content (community forum posts) is assumed to be factual
  - A “Climara patient” is defined as an author of a post or reply that makes a \*self-reference to having taken Climara
    - \* Identification of self-reference is done by LLM named-entity recognition prompt
  - “Sentiment” is categorized via LLM sentiment classifier prompt
- Solution
  - Performance Overview

# Guided Review Questions

(?) Who is the Climara patient in the community?

(?) How are community members feeling about Climara?

(?) What other treatments are members using?

## What is measured

9,991 documents · llama3.1:8b · prompt v7 · lexicon 46a8b905c4eb

Documents extracted 9,991 / 9,991

Schema-valid JSON returned 100.00%

**Named-entity recall — 34 entities, 6,813 pairs 95.6%**

the 34 of 113 lexicon entities with a single distinctive name that occurs in this corpus

Names in neither the document nor its parent 30

0.47% of the 6,320 mentions of scored entities; all 30 are one entity, black\_cohosh

Negation & demographic controls 8 / 8

Mentions mapped to a lexicon entity 88.5%

Mentions carrying a quoted context 98.5%

Surface forms collapsed to entities 3,159 → 110 (28.7 : 1)

Same 50 documents, extracted 3x — agreement 1.000 Jaccard

raw surface forms score 0.997 on the same sample — decoding is greedy at temperature=0

Produced by make evaluate and make stability.

# Insights and Findings

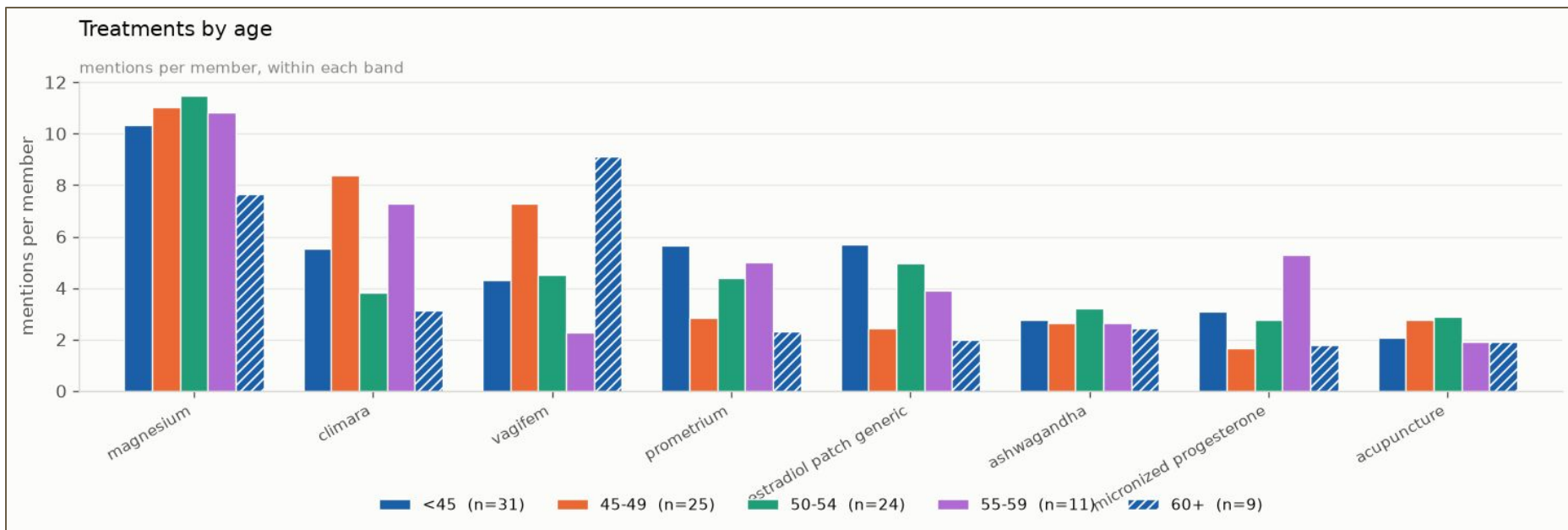
- Who is the MHT Menopause Patient Community?

|                                                                       |                                     |
|-----------------------------------------------------------------------|-------------------------------------|
| Members                                                               | 100                                 |
| Documents                                                             | 9,991 (2,001 posts · 7,990 replies) |
| Active                                                                | Oct 2025 – May 2026                 |
| Documents per member                                                  | 98 (74–121)                         |
| Age                                                                   | 49 (38–64)                          |
| Treatments named per member                                           | 20 (11–33)                          |
| Symptoms named per member                                             | 23 (15–29)                          |
| Switch events per member                                              | 3 (0–11)                            |
| The last five rows show <b>median (range)</b> across the 100 members. |                                     |

| Cut        | Group                    | Members |
|------------|--------------------------|---------|
| Age        | <45                      | 31      |
|            | 45–49                    | 25      |
|            | 50–54                    | 24      |
|            | 55–59                    | 11      |
|            | 60+                      | 9       |
| Location   | United States            | 52      |
|            | Canada                   | 21      |
|            | Australia                | 14      |
|            | UK & Ireland             | 12      |
| Profession | Corporate & professional | 41      |
|            | Healthcare               | 19      |
|            | Creative                 | 16      |
|            | Service & hospitality    | 13      |
|            | Education & other        | 9       |
| Climara    | Took Climara             | 51      |
|            | Did not take it          | 49      |

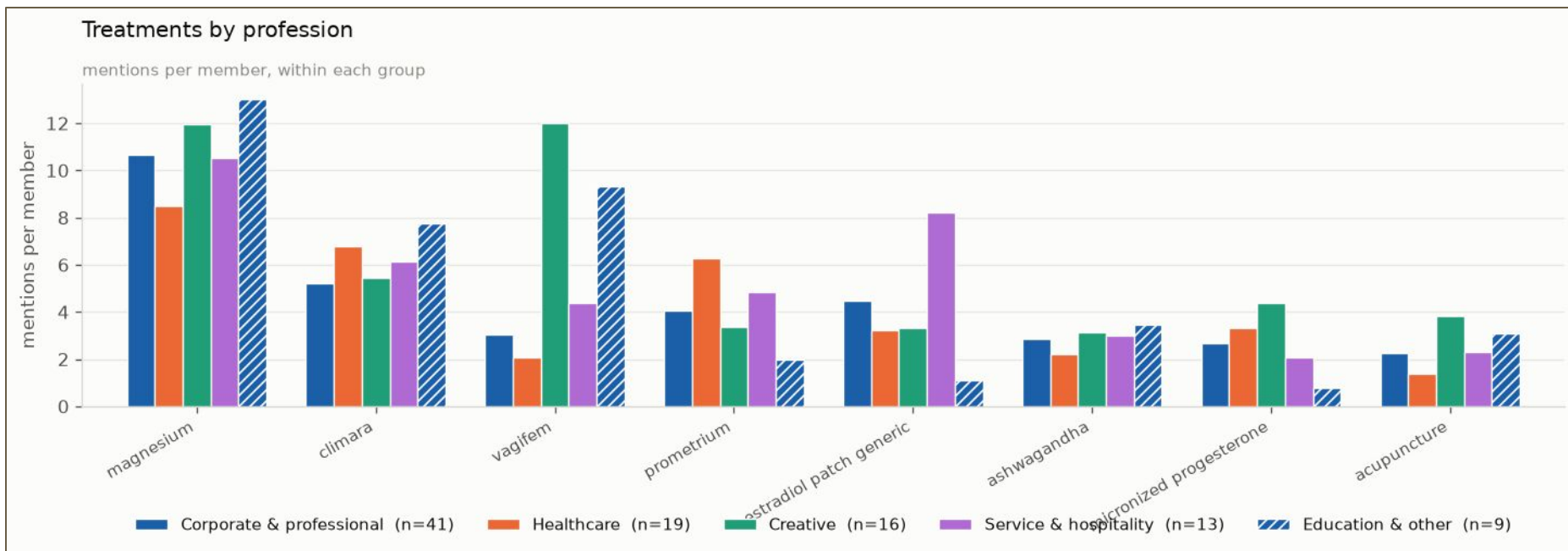
# Insights and Findings

- Who is the MHT Menopause Patient Community?



# Insights and Findings

- Who is the MHT Menopause Patient Community?

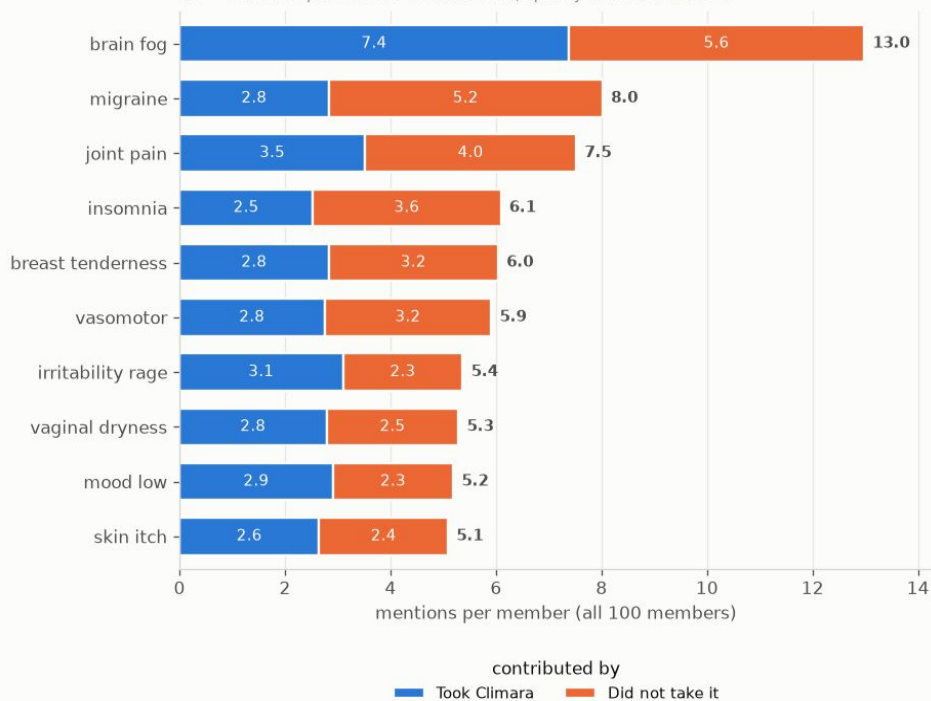


# Insights and Findings

- Who is the MHT Menopause - Climara Patient Community?

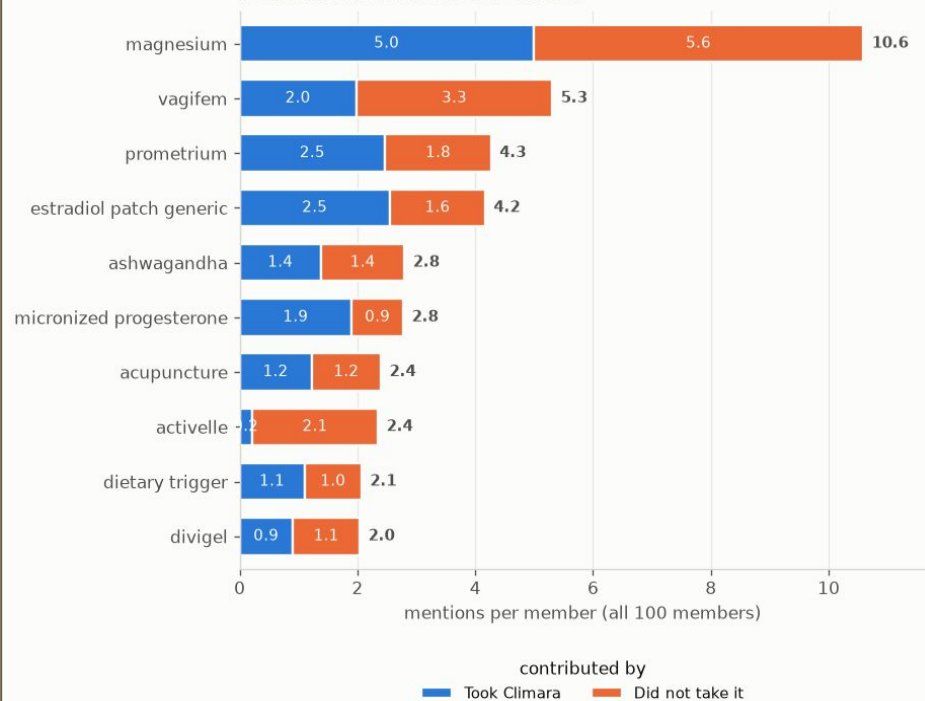
## What the community lives with

bar = mentions per member across all 100, split by who contributed it



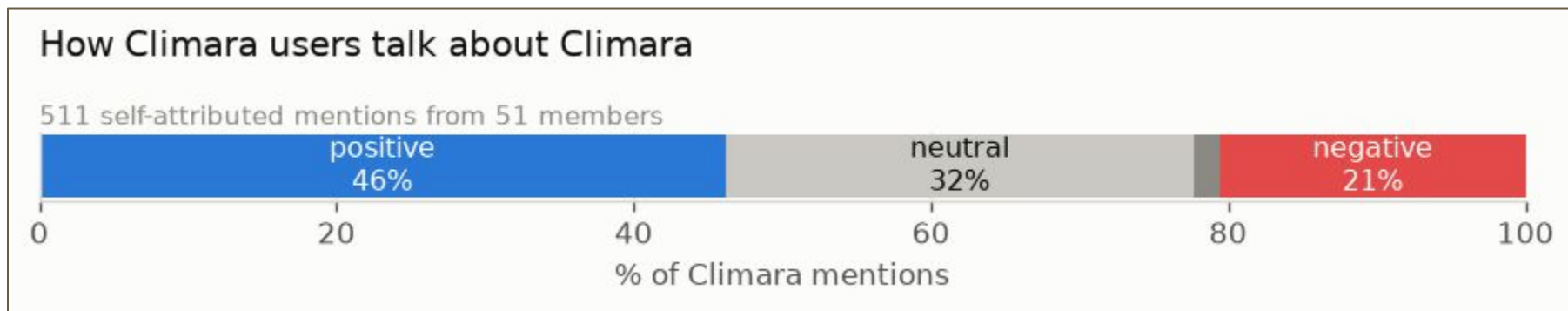
## What they take for it

umbrella entities and Climara itself excluded



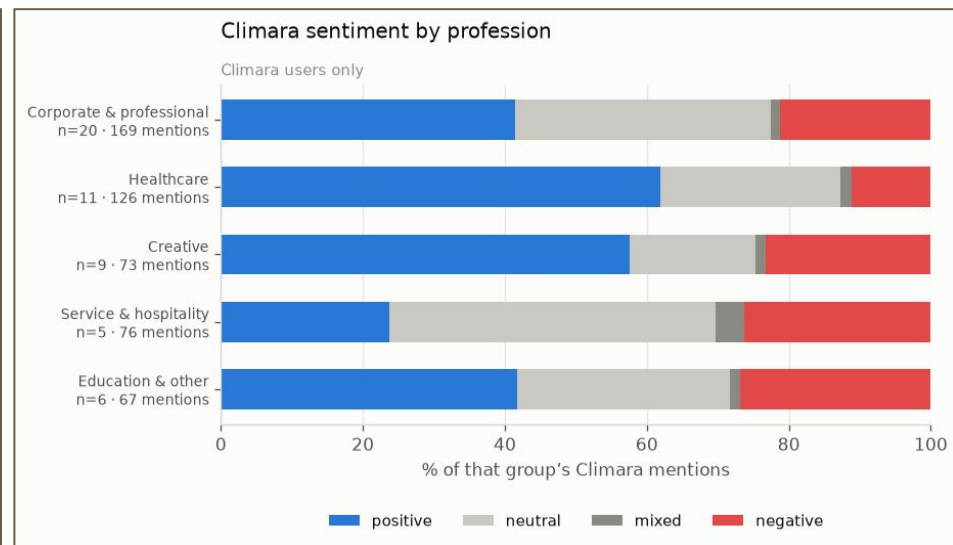
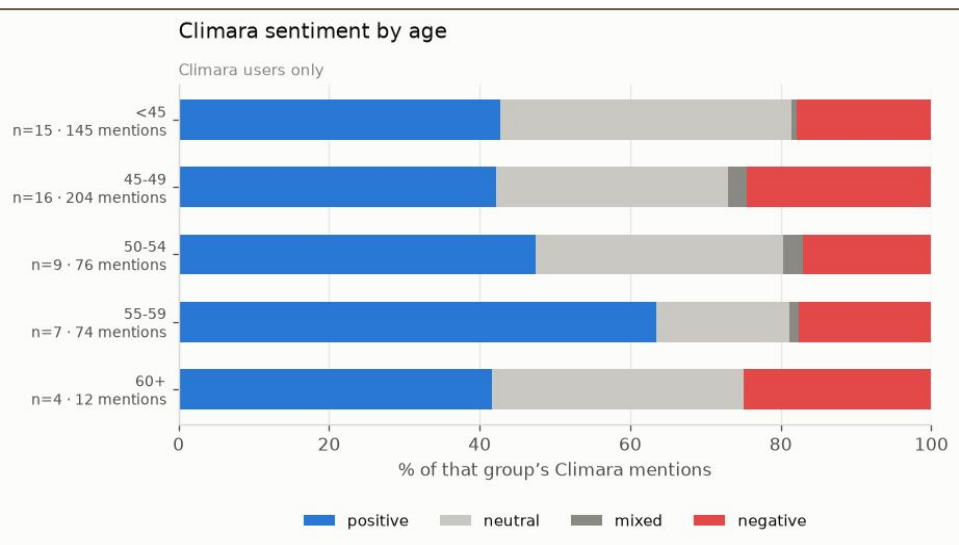
# Insights and Findings

- How does the MHT Climara Patient Community feel about the treatment?

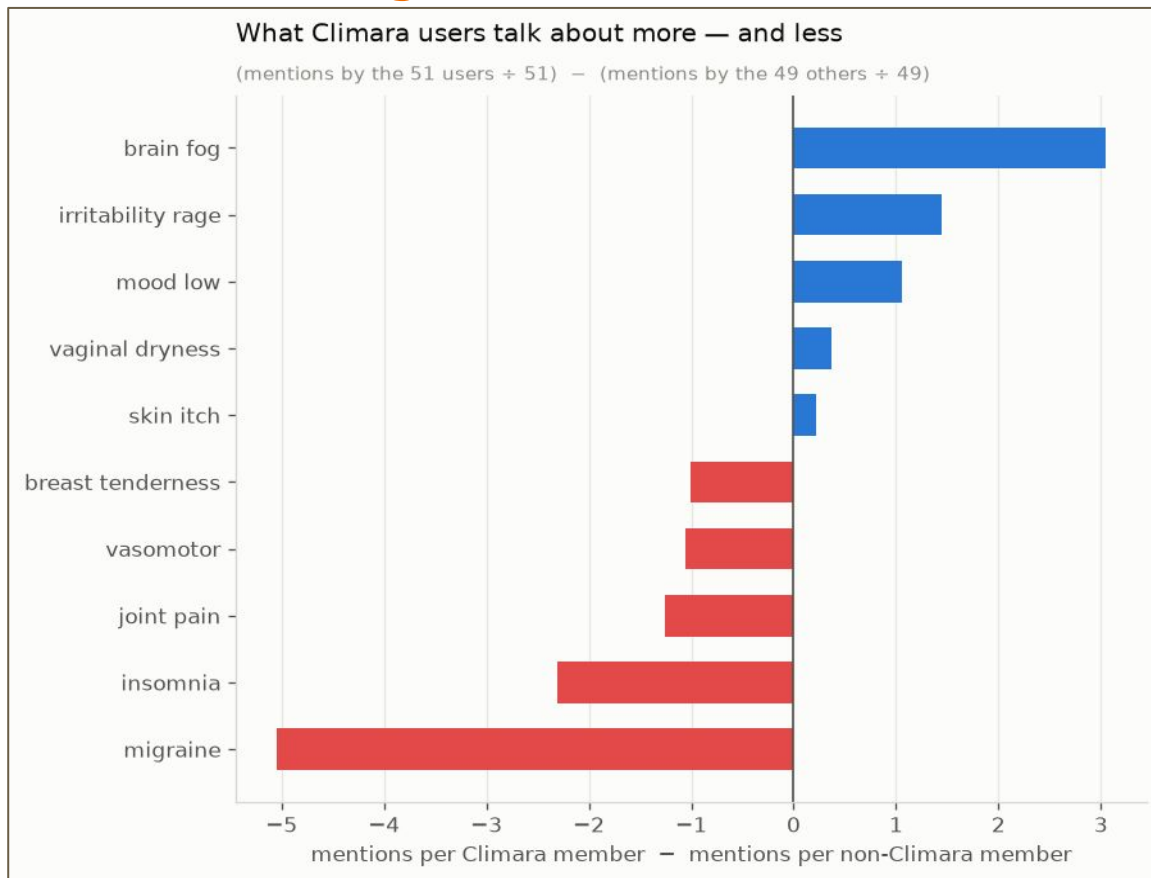


# Insights and Findings

- How does the MHT Climara Patient Community feel about the treatment?



# Insights and Findings



# Insights and Findings

- What treatments do the MHT Patient Community switch *TO* and *FROM*?

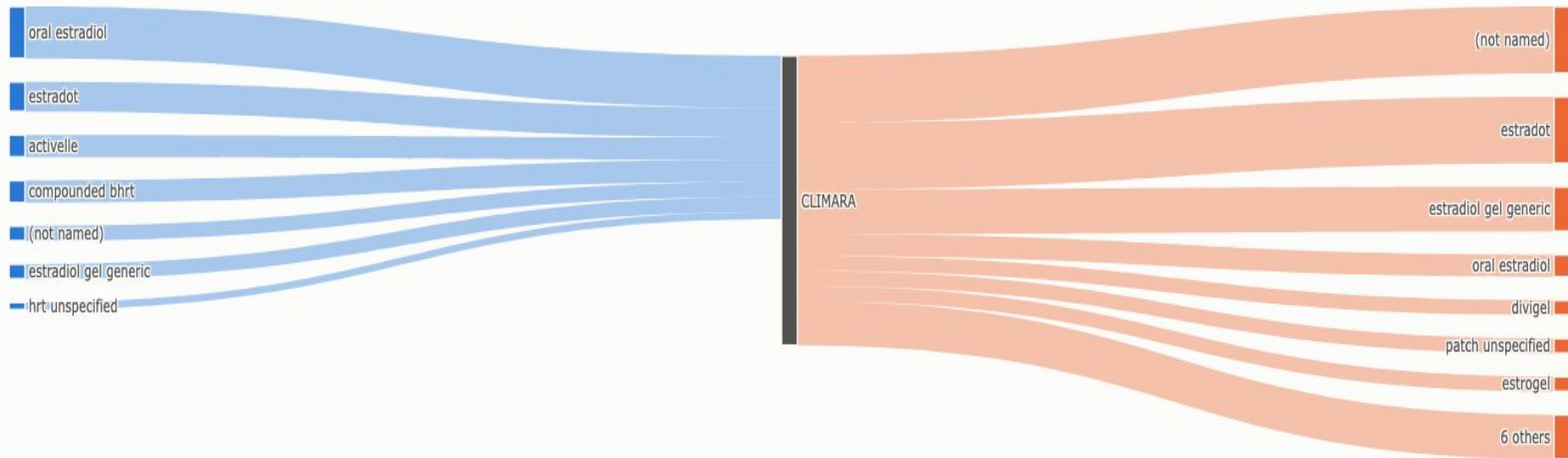
| Switches between treatment classes (159 of 338 events where both sides resolve to a class) |       |             |               |               |             |                 |            |            |
|--------------------------------------------------------------------------------------------|-------|-------------|---------------|---------------|-------------|-----------------|------------|------------|
| from ↓ to →                                                                                | Patch | Gel / spray | Oral estrogen | Oral combined | Progestogen | Local / vaginal | Local DHEA | Compounded |
| Patch                                                                                      | 17    | 55          | 4             | 1             | 3           | 2               |            | 1          |
| Gel / spray                                                                                | 9     | 1           |               |               |             | 2               |            |            |
| Oral estrogen                                                                              | 12    |             |               |               |             | 1               |            |            |
| Oral combined                                                                              | 10    |             |               |               | 1           | 2               |            |            |
| Progestogen                                                                                | 1     |             |               |               | 13          | 2               |            |            |
| Local / vaginal                                                                            | 3     | 1           | 1             |               |             | 7               | 2          |            |
| Compounded                                                                                 | 6     |             |               |               | 2           |                 |            |            |

# Insights and Findings

- What treatments do the MHT Climara Patient Community switch *TO* and *FROM*?

## Switching into and out of Climara

22 events in from 18 members - 39 out from 25 members



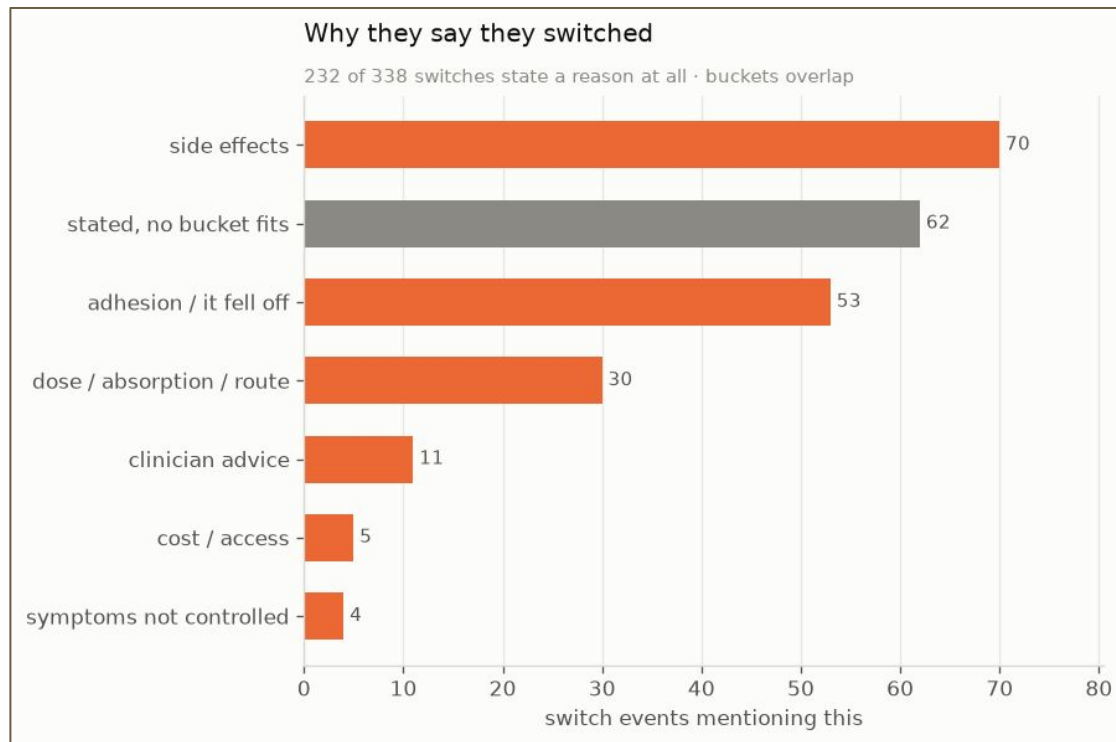
# Insights and Findings

- What treatments do the MHT Climara Patient Community switch *TO* and *FROM*?

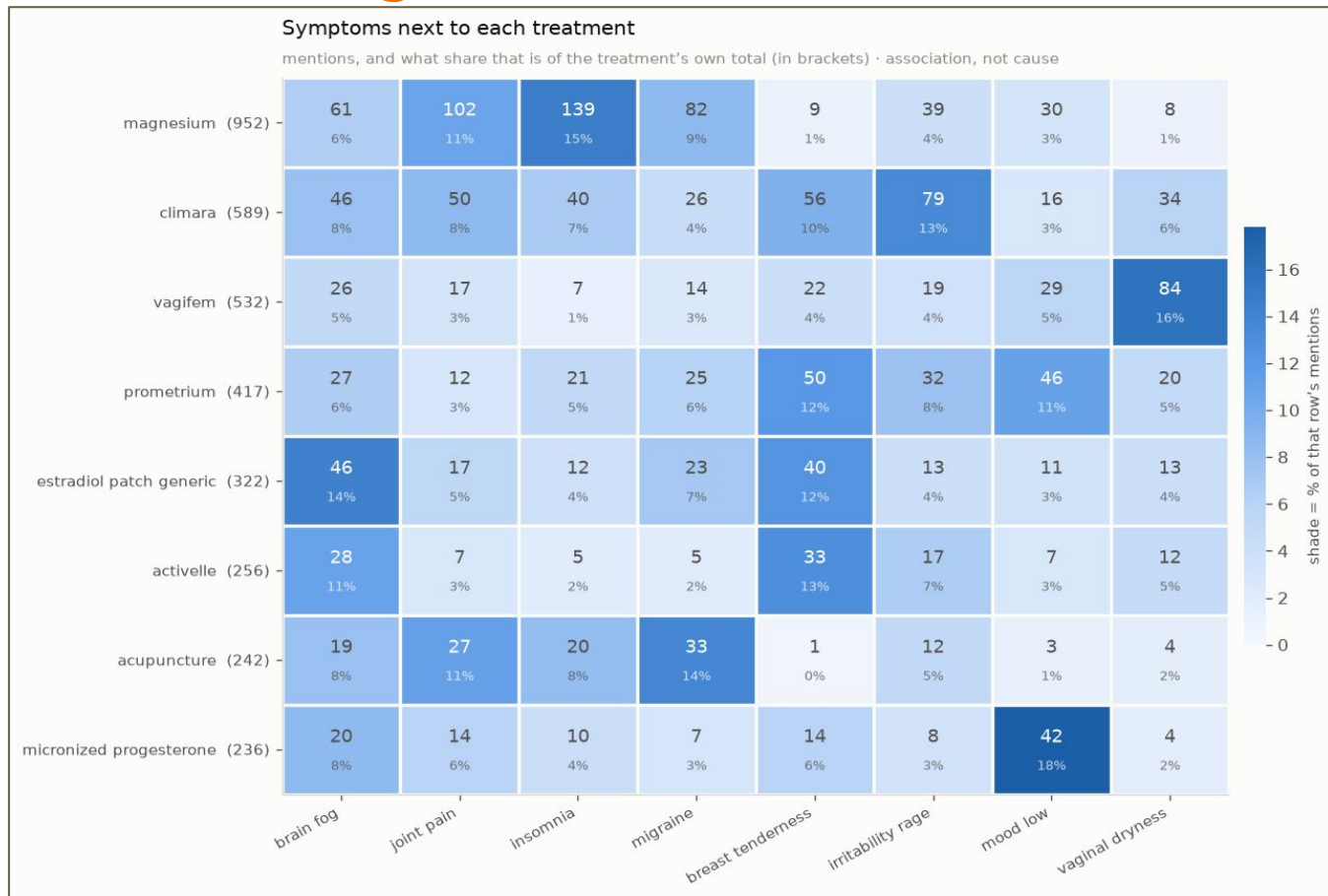
| Climara switch flows    |            |              |     |
|-------------------------|------------|--------------|-----|
| Treatment               | to Climara | from Climara | net |
| oral estradiol          | 7          | 3            | +4  |
| activelle               | 3          | 0            | +3  |
| compounded bhrt         | 3          | 0            | +3  |
| divigel                 | 0          | 2            | -2  |
| estrogel                | 0          | 2            | -2  |
| patch unspecified       | 0          | 2            | -2  |
| estradiol gel generic   | 2          | 6            | -4  |
| estradot                | 4          | 9            | -5  |
| (not named)             | 2          | 9            | -7  |
| 7 others, one move each | 1          | 6            | -5  |
| TOTAL                   | 22         | 39           | -17 |

# Insights and Findings

- Why does the MHT Menopause Patient Community switch treatments?



# Insights and Findings



# Additional Considerations & Next Steps

- Scaling for Production
  - Keep as LLM/agent-based → vLLM on cloud compute ⇒ 37 million posts for < ~\$5,000
- Validation
  - More human review if given more time
  - “lexicon” configuration vs. MHT Medical Ontology Reference database
  - “domain” configuration vs. MHT internal knowledge reference(s)
- Analytical Data Products
  - Switch Events - Refine prompt; more quality measures/tests
  -

# Q & A

Thank you for your time! What questions do we have?

# APPENDIX

# Sentiment - Member Comments

=== POSITIVE ===

- ...I'm not going to offer anything except – yes. This is grief and it's real and it deserves to be called what it is. I'm on the Climara patch now and it's helped the rage a bit but it hasn't fixed the feeling of being a stranger to myself, so I know that part isn't just...
- I made exactly this transition about two years ago – Vagifem plus added Climara patch, and then Prometrium. First month: breast tenderness yes, quite noticeable, lasted roughly six weeks for me. Bloating was a...
- ...me. Also did the iron supplementing cause any stomach issues? I'm already anxious about adding anything new on top of the Climara. Really glad you found something that works though, 1-2 nights vs 5-6 is massive.

=== NEGATIVE ===

- temple hair loss getting worse even on the patch – anyone else?? okay so i've been on the Climara patch for almost 8 months now and i genuinely thought it would help slow down the hair loss at my temples but honestly?? it seems like...
- ...exhausting, especially doing detailed design work through daily headaches💛 I stopped briefly about two years before starting my Climara patch and honestly the muscle aches were the thing that surprised me most – I didn't even realize HRT had been helping with them until...
- ...good evidence for meibomian gland dysfunction. I'm filing this away because my allergies have been flaring since starting the Climara patch and my eyes are part of it.

# Switch Events - Member Comments

22 of 338 switch events met the selection criteria; showing 5

SELECTION CRITERIA: Must carry a first-person movement construction, both endpoints must resolve to named products, and the surrounding text must not attribute the move to someone else

**vagifem** → **imvexxy** [applicator faff]

"Switched from Vagifem to Imvexxy a while back mostly because of applicator faff tbh"

**activelle** → **climara** [reduce first-pass hepatic effects]

"switching from oral to transdermal estrogen"

**estradot** → **estradiol gel generic** [thinking transdermal might be gentler]

"Switched to estrogen gel thinking transdermal might be gentler."

**prometrium** → **mirena**

"switching to a Mirena IUD for the progestogen component"

**vagifem** → **intrarosa** [anecdotally the urgency did settle a bit more]

"Switched to Intrarosa when my compounding clinic suggested it"

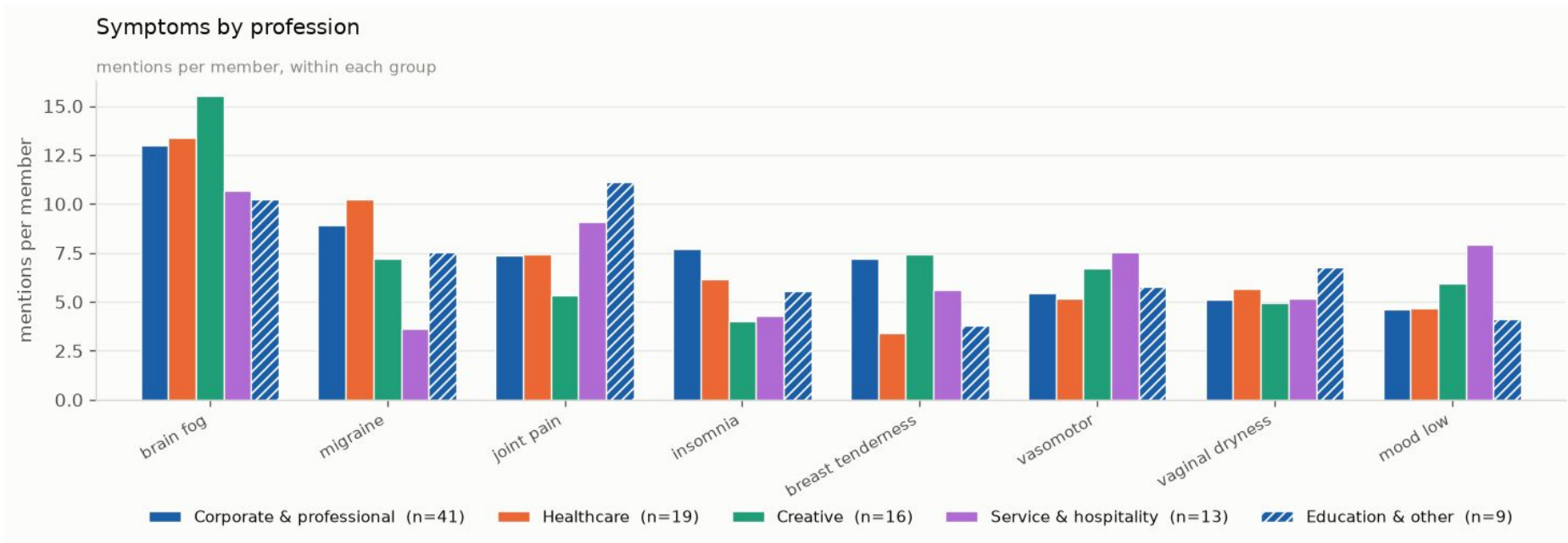
## User “switch event” - Reason Text Mapping

```
# The stated reasons, in members' own words. `reason` is free text with no
# canonical form - `adhesion issues` (47) and `adhesive issues` (6) are separate
# strings - so it is bucketed by regex rather than grouped. The patterns were
# widened against the actual unmatched values: efficacy is written as "breaking
# through", "not responding" and "wasn't resolving", never as "not working".
```

```
BUCKETS = {  
    'side effects': (r'tender|bleed|migraine|headache|nausea|bloat|irritat|rash|mood'  
        r'|adverse|side effect|ache|hair'),  
    'adhesion / it fell off': r'adhes|stick|peel|fall|fell off|came off|applicator',  
    'dose / absorption / route': (r'dose|absor|level|consistenc|strength|titrat|route'  
        r'|transdermal|systemic|peak|trough|hepatic'),  
    'clinician advice': r'doctor|gp|specialist|consultant|provider|nams|advice',  
    'cost / access': r'cost|price|insur|cover|afford|supply|shortage',  
    'symptoms not controlled': (r'not work|didn.t work|wasn.t work|no relief|ineffect'  
        r'|stopped work|nothing|breaking through|not respond'  
        r'|wasn.t respond|weren.t resolv|wasn.t resolv|got worse'  
        r'|unsustainable|wasn.t right'),  
}
```

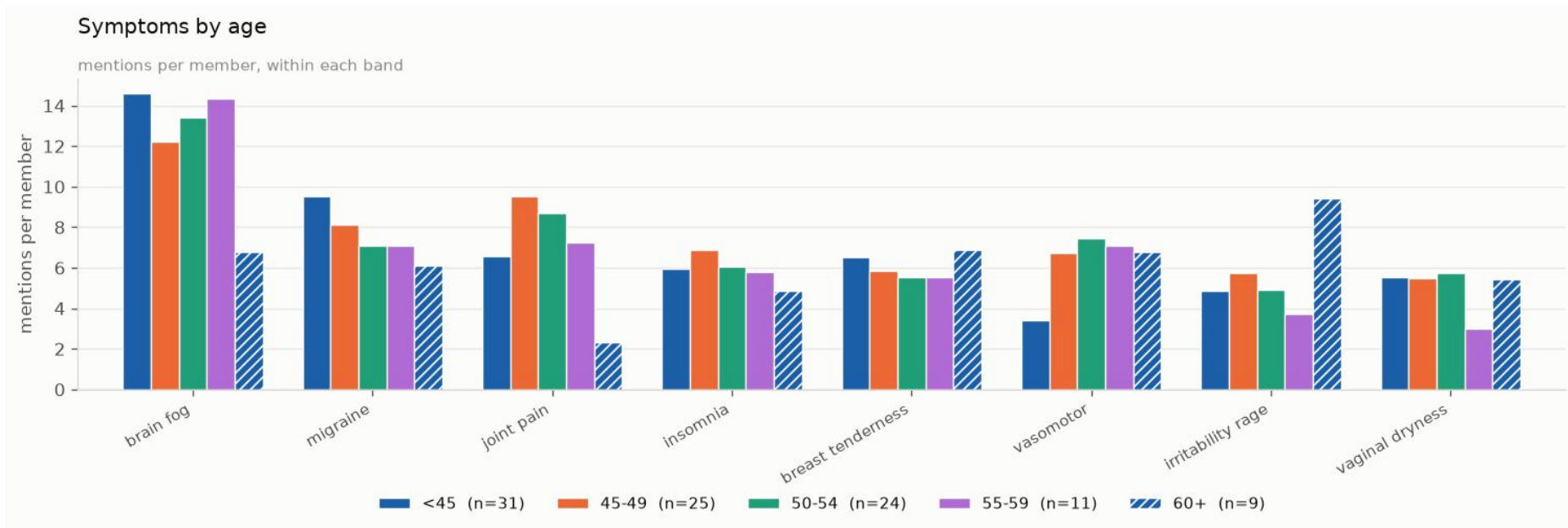
# Insights and Findings

- Who is the MHT Menopause Patient Community?



# Insights and Findings

- Who is the MHT Menopause Patient Community?



## How accurate is it?

Every figure below comes from one command — `make evaluate` — and needs no human annotation, so it re-runs in seconds on any version.

# 95.6%

**named-entity recall**

6,515 of 6,813 document-entity pairs

# 100%

**parse success**

9,991 of 9,991 documents · zero failures

# 8 / 8

**negation controls pass**

4 that must reject, 4 that must accept

# 0.47%

**names found nowhere in the text**

30 of 6,320 mentions of scored entities

### WHAT IS MEASURED

Documents extracted 9,991 / 9,991

Schema-valid JSON returned 100.00%

**Named-entity recall — 34 entities 95.6%**

Names in neither the document nor its parent 30

Negation & demographic controls 8 / 8

Mentions mapped to a lexicon entity 88.5%

Mentions carrying a quoted context 98.5%

Surface forms collapsed to entities 3,159 → 110 (28.7 : 1)

**Same 50 documents, extracted 3x — agreement on entities 1.000 Jaccard**

perfect. Greedy decoding at temperature=0 — determinism is a setting, not a hope.

Raw surface forms already score 0.997, so this measures determinism, not the value of canonicalisation — that case rests on the 28.7 : 1 collapse above.

#### Where it is weakest

Telehealth brands read as ordinary words in running text: `evernow` 71%, `mid_i_health` 71%.

All 30 unsupported names are one entity — `black_cohosh`, the known decoy case.

### HOW RECALL IS SCORED WITHOUT ANNOTATION

A **named** entity has no casual synonym. You either write “Climara” or you do not refer to it at all — unlike “the patch”, “my hormones” or “exercise”, which have a dozen phrasings.

#### So the gold standard is a string search:

for every document whose own text contains the name, did the extraction emit that entity? 6,813 pairs, zero labelling.

The entity list is derived from the lexicon, not hand-written, so swapping `lexicon.yml` for a different drug or disease carries the evaluation with it. Nothing here is specific to this corpus.

### WHAT IS NOT MEASURED

#### Three model-derived fields carry no accuracy number.

- `sentiment` — is “positive” about this treatment correct?
- `switch direction` — is from/to the right way round?
- `stance` — current vs past vs considering

Each needs human annotation. Recall and the controls do not, which is exactly why those are the numbers on this slide.

### REPRODUCE IT

`make evaluate`

Prints every figure above as a table. No model calls — it reads the stored extractions, so it is free and takes seconds.

llama3.1:8b · prompt v7 · lexicon 46a8b905c4eb  
9,991 documents · 5h45m · 0.48 docs/s on an M5 Max

The stability figure comes from `make stability`, which re-extracts a sample several times — the only check here that spends model calls (150 for the run above).